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Display device including a barrier layer to block light

Abstract

A display device may include: a display panel including a display area and a non-display area, the display area having a light emitting area and a non-light emitting area; at least one first barrier layer on the non-light emitting area of the display panel, the at least one first barrier layer configured to block light emitted from the light emitting area; at least one second barrier layer overlapping the at least one first barrier layer in the non-emitting area, the at least one second barrier layer configured to block light emitted from the light emitting area; and at least one lens overlapping the light emitting area and on a same layer as the at least one first barrier layer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION
(1) The present application claims the priority benefit of Republic of Korea Patent Application No. 10-2022-0109898 filed in Republic of Korea on Aug. 31, 2022, which is incorporated by reference in its entirety.
TECHNICAL FIELD
(2) The present disclosure relates to a display device, and more particularly, to a display device

where a viewing angle is adjusted or cut off without a light control film.

DISCUSSION OF THE RELATED ART

(3) As a technology for a display device progresses, display devices of various types such as a liquid crystal display (LCD) device and an organic light emitting diode (OLED) display device have been developed.

(4) A display device sensing a touch as well as displaying an image has been widely used, and a display device having a touch sensing layer has been developed.

(5) A display device is used for a vehicle such as a car to provide various information. For example, to prevent obstruction of a view during driving due to reflection of an image of the display device on a windscreen of a car, an additional light control film that controls or cuts off a viewing angle by blocking light emitted along a vertical direction is attached to a display panel of the display device.

(6) However, a luminance of the display device is reduced and a touch sensitivity is deteriorated due to the light control film.

SUMMARY

(7) Accordingly, embodiments of the present disclosure are directed to a display device that substantially obviate one or more of problems due to the limitations and disadvantages of the related art.

(8) The inventors have invented a display device that controls or cuts off a viewing angle without attaching a light control film on a display panel.

(9) An object of the present disclosure is to provide a display device where a viewing angle is controlled or cut off and a front luminance is improved through an increase of a light collection efficiency.

(10) Another object of the present disclosure is to provide a display device where a viewing angle is controlled or cut off without a decrease of a touch sensitivity.

(11) In one embodiment, a display device comprises: a display panel including a display area and a non-display area, the display area having a light emitting area and a non-light emitting area; at least one first barrier layer on the non-light emitting area of the display panel, the at least one first barrier layer configured to block light emitted from the light emitting area; at least one second barrier layer overlapping the at least one first barrier layer in the non-emitting area, the at least one second barrier layer configured to block light emitted from the light emitting area; and at least one lens overlapping the light emitting area and on a same layer as the at least one first barrier layer.

(12) In one embodiment, a display device comprises: a substrate including a display area and a non-display area; a plurality of emitting elements in the display area, the plurality of emitting elements configured to emit light; at least one encapsulating layer on the plurality of emitting elements; at least one barrier layer on the at least one encapsulating layer and non-overlapping with an emitting element from the plurality of emitting elements, the at least one barrier layer configured to block the light emitted by the plurality of emitting elements; and a plurality of lenses on the at least one encapsulating layer, the plurality of lenses overlapping the plurality of emitting elements.

(13) In one embodiment, a display device comprises: a substrate including a display area and a non-display area; a plurality of light emitting elements in the display area, the plurality of light emitting elements configured to emit light; one or more intermediate layers on the plurality of light emitting elements; a plurality of touch electrodes on the one or more intermediate layers and non-overlapping with the plurality of light emitting elements, the plurality of touch electrodes configured to sense touch and block the light emitted from the plurality of light emitting elements; and a plurality of lenses that are overlapping with the plurality of light emitting elements.

(14) Additional features and advantages of the disclosure will be set forth in the description which follows, and in part will be apparent to those skilled in the art from the description or may be learned by practice of the disclosure. These and other advantages of the disclosure may be realized and attained by the structure particularly pointed out in, or derivable from, the written description, claims hereof, and the appended drawings.

(15) It is to be understood that both the foregoing general description and the following detailed description are explanatory and by way of examples and are intended to provide further explanation of the disclosure as claimed without limiting its scope.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification, illustrate embodiments of the disclosure and together with the description serve to explain the principles of the disclosure. In the drawings:

(2) FIG. **1** is view showing an interior of a vehicle having a display device according to a first embodiment of the present disclosure;

(3) FIG. **2** is a perspective view showing a display device according to the first embodiment of the present disclosure;

(4) FIG. **3** is a block diagram showing a display device according to the first embodiment of the present disclosure;

(5) FIG. **4** is a plan view showing a display device according to the first embodiment of the present disclosure;

(6) FIG. **5** is a plan view showing a touch sensing layer of a display device according to the first embodiment of the present disclosure;

(7) FIG. **6** is a magnified plan view showing a touch electrode of a display device according to the first embodiment of the present disclosure;

(8) FIG. **7** is a cross-sectional view taken along lines I-I′ and II-II′ of FIGS. **4** and **5** according to the first embodiment of the present disclosure;

(9) FIG. **8** is a magnified cross-sectional view of a region **7**A of FIG. **7** according to the first embodiment of the present disclosure;

(10) FIG. **9** is a magnified cross-sectional view showing a lens of FIG. **7** according to the first embodiment of the present disclosure;

(11) FIG. **10** is a graph showing a normalized transmittance intensity with respect to an up and down viewing angle of a display device according to the first embodiment of the present disclosure;

(12) FIG. **11** is a cross-sectional view showing a display device according to a second embodiment of the present disclosure;

(13) FIG. **12** is a magnified cross-sectional view of a region **11**A of FIG. **11** according to the second embodiment of the present disclosure;

(14) FIG. **13** is a graph showing a normalized transmittance intensity with respect to an up and down viewing angle of a display device according to the second embodiment of the present disclosure;

(15) FIG. **14** is a cross-sectional view showing a display device according to a third embodiment of the present disclosure;

(16) FIG. **15** is a magnified cross-sectional view of a region **14**A of FIG. **14** according to the third embodiment of the present disclosure;

(17) FIG. **16** is a graph showing a normalized transmittance intensity with respect to an up and down viewing angle of a display device according to the third embodiment of the present disclosure;

(18) FIG. **17** is a cross-sectional view showing a display device according to a fourth embodiment of the present disclosure; and

(19) FIG. **18** is a cross-sectional view showing a display device according to a fifth embodiment of

the present disclosure.

DETAILED DESCRIPTION

(20) Advantages and features of the present disclosure, and implementation methods thereof will be clarified through following example embodiments described with reference to the accompanying drawings. The present disclosure may, however, be embodied in different forms and should not be construed as limited to the example embodiments set forth herein. Rather, these example embodiments are provided so that this disclosure may be sufficiently thorough and complete to assist those skilled in the art to fully understand the scope of the present disclosure. Further, the protected scope of the present disclosure is defined by claims and their equivalents.

(21) The shapes, sizes, ratios, angles, numbers, and the like, which are illustrated in the drawings to describe various example embodiments of the present disclosure, are merely given by way of example. Therefore, the present disclosure is not limited to the illustrations in the drawings. Like reference numerals refer to like elements throughout, unless otherwise specified.

(22) In the following description, where the detailed description of the relevant known function or configuration may unnecessarily obscure a feature or aspect of the present disclosure, a detailed description of such known function or configuration may be omitted or a brief description may be provided.

(23) Where the terms “comprise,” “have,” “include,” and the like are used, one or more other elements may be added unless the term, such as “only,” is used. An element described in the singular form is intended to include a plurality of elements, and vice versa, unless the context clearly indicates otherwise.

(24) In construing an element, the element is to be construed as including an error or tolerance range even where no explicit description of such an error or tolerance range is provided.

(25) Where positional relationships are described, for example, where the positional relationship between two parts is described using “on,” “over,” “under,” “above,” “below,” “beside,” “next,” or the like, one or more other parts may be located between the two parts unless a more limiting term, such as “immediate(ly),” “direct(ly),” or “close(ly)” is used. For example, where an element or layer is disposed “on” another element or layer, a third layer or element may be interposed therebetween.

(26) Although the terms “first”, “second”, “A”, “B”, “(a)”, “(b)”, and the like may be used herein to refer to various elements, these elements should not be interpreted to be limited by these terms as they are not used to define a particular order or precedence. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure.

(27) Features of various embodiments of the present disclosure may be partially or entirely coupled to or combined with each other. They may be linked and operated technically in various ways as those skilled in the art can sufficiently understand. The embodiments may be carried out independently of or in association with each other in various combinations.

(28) Hereinafter, a display device according to various example embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

(29) FIG. **1** is view showing an interior of a vehicle having a display device according to a first embodiment of the present disclosure, FIG. **2** is a perspective view showing a display device according to a first embodiment of the present disclosure, and FIG. **3** is a block diagram showing a display device according to a first embodiment of the present disclosure.

(30) In FIG. **1**, a display device **100** according to a first embodiment of the present disclosure may be disposed on a portion of a dash board of a vehicle. The dash board may be disposed in front of a front seat (e.g., driver seat, passenger seat) of the vehicle. For example, the dashboard may include an input unit for manipulating various equipment (e.g., air conditioner, audio system, navigation system).

(31) The display device **100** on the dash board may be used as an input unit for manipulating a portion of the various equipment of the vehicle. The display device **100** may provide various information regarding the vehicle, for example, a driving information (e.g., present speed, residual fuel quantity, mileage) and a part information (e.g., damage degree of tire).
(32) The display device **100** may be disposed to extend across the driver seat and the passenger seat in the front seat of the vehicle. A user of the display device **100** may include a driver of the vehicle and a fellow passenger in the passenger seat. Both of the driver and the fellow passenger may use the display device **100**.
(33) FIG. **1** may show a portion of the display device **100**. FIG. **1** may show a display panel among a plurality of elements of the display device **100**. The other elements of the display device **100** may be disposed in the interior (or a portion) of the vehicle.
(34) In FIGS. **2** and **3**, the display device **100** according to the first embodiment of the present disclosure includes a display panel **110**, a scan driving unit **120** (e.g., a circuit), a timing controlling unit **160** (e.g., a circuit), a host system **170**, a touch driving unit **180** (e.g., a circuit), and a touch coordinate calculating unit **190** (e.g., a circuit).
(35) Although the display device **100** is exemplarily illustrated as an organic light emitting diode (OLED) display device in a first embodiment of the present disclosure, the display device **100** may be various display devices such as a liquid crystal display (LCD) device in another embodiment.
(36) The display panel **110** may include a first substrate **111**, a second substrate **112** and a thin film transistor layer, an emitting element layer, an encapsulating layer and a touch sensing layer between the first and second substrates **111** and **112**. The embodiments of the present disclosure are not limited thereto. The second substrate **112** may include a plastic film, a glass substrate or an encapsulation film.
(37) The display panel **110** may include a display area where a plurality of subpixels P display an image. A plurality of data lines D**1** to Dm (m is a positive integer greater than or equal to 2) and a plurality of scan lines S**1** to Sn (n is a positive integer greater than or equal to 2) are formed in the display panel **110**. The plurality of data lines D**1** to Dm and the plurality of scan lines S**1** to Sn may cross each other. A scan line may be referred to as a gate line. The subpixel P may be disposed in a region defined by crossing of the scan line and the data line.
(38) Each of the plurality of subpixels P of the display panel **110** may be connected to one of the plurality of data lines D**1** to Dm and one of the plurality of scan lines S**1** to Sn.
(39) Each of the plurality of subpixels P of the display panel **110** may include a driving transistor adjusting a current between a source electrode and a drain electrode according to a data voltage applied to a gate electrode, a scan transistor turned on according to a scan signal of the scan line and supplying the data voltage of the data line to the gate electrode of the driving transistor, a light emitting diode emitting a light according to the current between the source electrode and the drain electrode of the driving transistor and a storage capacitor storing a voltage of the gate electrode of the driving transistor. As a result, each subpixel P may emit a light according to a current supplied to the light emitting diode.
(40) The scan driving unit **120** receives a scan control signal GCS from the timing controlling unit **160**. The scan driving unit **120** supplies the scan signal to the plurality of scan lines S**1** to Sn according to the scan control signal GCS.
(41) The scan driving unit **120** may be disposed in a non-display area at one side or both sides of a display area of the display panel **110** as a gate in panel (GIP) type. Alternatively, the scan driving unit **120** may be formed as a driving chip mounted on a flexible film and may be attached to the non-display area at one side or both sides of a display area of the display panel **110** as a tape automated bonding (TAB) type.
(42) The data driving unit **130** receives a digital video data DATA and a data control signal DCS from the timing controlling unit **160**. The data driving unit **130** converts the digital video data DATA into an analog data voltage of positive and negative polarities according to the data control

signal DCS and supplies the analog data voltage to the plurality of data lines D**1** to Dm. For example, the subpixel P where the data voltage is supplied is selected according to the scan signal of the scan driving unit **120** and the data voltage is supplied to the selected subpixel P.

(43) The data driving unit **130** may include a plurality of source drive integrated circuits (ICs) **131**. Each of the plurality of source drive ICs **131** may be mounted on a flexible film **140** as a chip on film (COF) type or a chip on plastic (COP) type. The flexible film **140** may be attached to pads in the non-display area of the display panel **110** using an anisotropic conductive film (ACF) such that the plurality of source drive ICs **131** are connected to the pads.

(44) A circuit board **150** may be attached to the flexible film **140**. A plurality of circuits as driving chips may be mounted on the circuit board **150**. For example, the timing controlling unit **160** may be mounted on the circuit board **150**. The circuit board **150** may include a printed circuit board (PCB) or a flexible printed circuit board.

(45) The timing controlling unit **160** receives the digital video data DATA and a plurality of timing signals from the host system **170**. The plurality of timing signals may include a vertical synchronization signal, a horizontal synchronization signal, a data enable signal and a dot clock. The vertical synchronization signal is a signal defining one frame period. The horizontal synchronization signal is a signal defining one horizontal period required for supplying the data voltage to the subpixels of one horizontal line of the display panel **110**. The data enable signal is a signal defining a period where an effective data is inputted. The dot clock is a repeated signal with a relatively short period.

(46) To control an operation timing of the scan driving unit **120** and the data driving unit **130**, the timing controlling unit **160** generates the data control signal DCS for controlling the operation timing of the data driving unit **130** and the scan control signal GCS for controlling the operation timing of the scan driving unit **120** based on the plurality of timing signals. The timing controlling unit **160** outputs the scan control signal GCS to the scan driving unit **120** and outputs the digital video data DATA and the data control signal DCS to the data driving unit **130**.

(47) The host system **170** may include a navigation system, a set-top box, a DVD player, a Blu-ray player, a personal computer (PC), a home theater system, a broadcast receiver and a phone system. The host system **170** includes a system on chip (SoC) having a scaler and converts the digital video data DATA of an input image into a format suitable for the display panel **110**. The host system **170** transmits the digital video data DATA and the plurality of timing signals to the timing controlling unit **160**.

(48) A plurality of first touch electrodes and a plurality of second touch electrodes as well as the plurality of data lines D**1** to Dm and the plurality of scan lines S**1** to Sn may be formed in the display panel **110**. The plurality of first touch electrodes may cross the plurality of second touch electrodes. The plurality of first touch electrodes may be connected to a first touch driving unit **181** through a plurality of first touch lines T**1** to Tj (j is a positive integer greater than or equal to 2). The plurality of second touch electrodes may be connected to a second touch driving unit **182** through a plurality of second touch lines R**1** to Ri (i is a positive integer greater than or equal to 2). A touch sensor may be formed at each crossing of the plurality of first touch electrodes and the plurality of second touch electrodes. Although the touch sensor exemplarily has a mutual capacitance type in an embodiment of the present disclosure, it is not limited thereto.

(49) The touch driving unit **180** supplies a driving pulse to the plurality of first touch electrodes through the plurality of first touch lines T**1** to Tj and senses a charge change amount of each touch sensor through the plurality of second touch lines R**1** to Ri. In FIG. **3**, the plurality of first touch lines T**1** to Tj are transmitting (Tx) lines supplying the driving pulse, and the plurality of second touch lines R**1** to Ri are receiving (Rx) lines sensing the charge change amount of each touch sensor.

(50) The touch driving unit **180** includes the first touch driving unit **181** (e.g., a circuit), the second touch driving unit **182** (e.g., a circuit) and a touch controlling unit **183** (e.g., a circuit). The first

touch driving unit **181**, the second touch driving unit **182** and the touch controlling unit **183** may be formed in a single read out integrated circuit (ROIC).
(51) The first touch driving unit **181** selects a first touch line where the driving pulse is outputted according to a control of the touch controlling unit **183** and supplies the driving pulse to the selected first touch line. For example, the first touch driving unit **181** may sequentially supply the driving pulse to the plurality of first touch lines T**1** to Tj.
(52) The second touch driving unit **182** selects a second touch line where the charge change amount of the touch sensors is received according to a control of the touch controlling unit **183** and receives the charge change amount from the selected second touch line. The second touch driving unit **182** selects (sampling) the charge change amount of the touch sensors received through the plurality of second touch lines R**1** to Ri and converts the selected charge change amount into a touch raw data TRD of a digital type.
(53) The touch controlling unit **183** may generate a transmitting (Tx) setup signal for selecting the first touch line where the driving pulse is outputted from the first touch driving unit **181** and a receiving (Rx) setup signal for selecting the second touch line where a touch sensor voltage is received by the second touch driving unit **182**. In addition, the touch controlling unit **183** may generate a plurality of timing control signals for controlling an operation timing of the first touch driving unit **181** and the second touch driving unit **182**.
(54) The touch coordinate calculating unit **190** receives the touch raw data TRD from the touch driving unit **180**. The touch coordinate calculating unit **190** calculates touch coordinates according to a touch coordinate calculating method and outputs a touch coordinate data HIDxy including an information of the touch coordinates to the host system **170**.
(55) The touch coordinate calculating unit **190** may be formed as a micro controller unit (MCU). The host system **170** performs an application program relating to a coordinate where a touch occurs by a user by analyzing the touch coordinate data HIDxy inputted from the touch coordinate calculating unit **190**. The host system **170** transmits the digital video data DATA and the plurality of timing signals to the timing controlling unit **160**.
(56) The touch driving unit **180** may be formed in the source drive ICs **131** or a separate driving chip to be mounted on the circuit board **150**. In addition, the touch coordinate calculating unit **190** may be formed in a driving chip to be mounted on the circuit board **150**.
(57) The display device according to a first embodiment of the present disclosure may be illustrated with reference to FIGS. **4** to **8** hereinafter.
(58) FIG. **4** is a plan view showing a display device according to the first embodiment of the present disclosure, FIG. **5** is a plan view showing a touch sensing layer of a display device according to the first embodiment of the present disclosure, FIG. **6** is a magnified plan view showing a touch electrode of a display device according to the first embodiment of the present disclosure, FIG. **7** is a cross-sectional view taken along lines I-I′ and II-II′ of FIGS. **4** and **5** according to the first embodiment of the present disclosure, and FIG. **8** is a magnified cross-sectional view of a region **7**A of FIG. **7** according to the first embodiment of the present disclosure.
(59) In FIGS. **4** to **8**, a first substrate **111** may include a display area DA and a non-display area NDA. The first substrate **111** may be a glass substrate or a plastic substrate having a flexible material such as polyimide. The embodiments of the present disclosure are not limited thereto.
(60) The non-display area NDA may include a pad area PA where pads PAD are disposed and a dam DAM is disposed in the non-display area NDA on the first substrate **111**. The dam DAM may be formed to have a plurality of dams and the plurality of dams may be formed in different layers.
(61) A thin film transistor layer and an emitting element layer (e.g., a light emitting layer) are disposed in the display area DA on the first substrate **111**.
(62) The thin film transistor layer includes a plurality of thin film transistors **210**, a gate insulating layer **220**, an interlayer insulating layer **230**, a passivation layer **240** and a first planarizing layer **250**.

(63) A buffer layer **113** may be disposed on the first substrate **111**. The buffer layer **113** is formed on the first substrate **111** to protect the plurality of thin film transistors **210** and a plurality of emitting elements **260** (e.g., light emitting elements) from moisture penetrating through the first substrate **111**. The first substrate **111** may face into a second substrate **112**. The buffer layer **113** may include a plurality of inorganic layers alternately laminated. For example, the buffer layer **113** may have a multiple layer including a plurality of inorganic layers of at least one of silicon oxide (SiOx), silicon nitride (SiNx) and silicon oxynitride (SiON).
(64) A thin film transistor **210** is disposed on the buffer layer **113**. The thin film transistor **210** includes an active layer **211**, a gate electrode **212**, a source electrode **213** and a drain electrode **214**. Although the thin film transistor **210** exemplarily have a top gate type where the gate electrode **212** is disposed on the active layer **211**, it is not limited thereto. The thin film transistor **210** may have a bottom gate type or a double gate type in another embodiment.
(65) The active layer **211** is disposed on the buffer layer **113**. Although the active layer **211** may include an oxide semiconductor material such as indium gallium zinc oxide (IGZO), it is not limited thereto. The active layer **211** may include low temperature polycrystalline silicon (LTPS) or amorphous silicon (a-Si) in another embodiment.
(66) A light shielding layer (not shown) for blocking an external light incident to the active layer **211** may be disposed between the buffer layer **113** and the active layer **211**.
(67) A gate insulating layer **220** insulating the active layer **211** and the gate electrode **212** may be disposed on the active layer **211**. Although the gate insulating layer **220** is exemplarily disposed on the entire first substrate **111**, it is not limited thereto. The gate insulating layer **220** may be disposed only under the gate electrode **212**. The gate insulating layer **220** may have an inorganic layer, for example, a single layer or a multiple layer of silicon oxide (SiOx) and silicon nitride (SiNx). The embodiments of the present disclosure are not limited thereto.
(68) The gate electrode **212** and a gate line (not shown) may be disposed on the gate insulating layer **220**. The gate electrode **212** and the gate line may have a single layer or a multiple layer of one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) and copper (Cu) and an alloy thereof. The embodiments of the present disclosure are not limited thereto.
(69) The interlayer insulating layer **230** may be disposed on the gate electrode **212** and the gate line. The interlayer insulating layer **230** may have an inorganic layer, for example, a single layer or a multiple layer of silicon oxide (SiOx) and silicon nitride (SiNx). The embodiments of the present disclosure are not limited thereto.
(70) The source electrode **213**, the drain electrode **214** and a data line (not shown) may be disposed on the interlayer insulating layer **230**. The source electrode **213** and the drain electrode **214** may be connected to the active layer **211** through contact holes in the gate insulating layer **220** and the interlayer insulating layer **230**. In one embodiment, the source electrode **213** and the drain electrode **214** are made of a same material as the pads PAD. For example, the pads PAD include the same material as the source electrode **213** and the drain electrode **214**. In one embodiment, the pads PAD include the same material as the gate electrode **212**. The source electrode **213**, the drain electrode **214** and the data line may have a single layer or a multiple layer of one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) and copper (Cu) and an alloy thereof. The embodiments of the present disclosure are not limited thereto.
(71) The passivation layer **240** for insulating the thin film transistor **210** may be disposed on the source electrode **213**, the drain electrode **214** and the data line. The passivation layer **240** may have an inorganic layer, for example, a single layer or a multiple layer of silicon oxide (SiOx) and silicon nitride (SiNx). The embodiments of the present disclosure are not limited thereto.
(72) The first planarizing layer **250** may be disposed on the passivation layer **240** to planarize a step difference due to the thin film transistor **210**. The first planarizing layer **250** may include an organic material such as acrylic resin, epoxy resin, phenolic resin, polyamide resin and polyimide

resin. The embodiments of the present disclosure are not limited thereto.
(73) The emitting element layer is disposed on the thin film transistor layer. The emitting element layer includes a plurality of emitting elements **260** and a bank layer **270**.
(74) For example, the plurality of emitting elements **260** may be disposed in the display area DA. In one embodiment, the emitting elements **260** are micro-LEDs or organic light emitting diodes (OLED).
(75) The plurality of emitting elements **260** and the bank layer **270** are disposed on the first planarizing layer **250**. The emitting element **260** of a light emitting diode includes a first electrode **261**, an emitting layer **262** (e.g., a light emitting layer) and a second electrode **263**. The first electrode **261** may be an anode and the second electrode **263** may be a cathode.
(76) The first electrode **261** may be disposed on the first planarizing layer **250**. The first electrode **261** is connected to the source electrode **213** of the thin film transistor **210** through a contact hole in the passivation layer **240** and the first planarizing layer **250**. The first electrode **261** may include a metallic material having a relatively high reflectance and may have a laminated structure such as Ti/Al/Ti of aluminum and titanium, ITO/Al/ITO of aluminum and indium tin oxide (ITO), a silver palladium copper (AgPdCu:APC) alloy and ITO/APC/ITO of APC alloy and ITO. The embodiments of the present disclosure are not limited thereto.
(77) The bank layer **270** may be disposed on the first planarizing layer **250** to cover an edge portion of the first electrode **261** for defining the subpixel P. For example, the bank layer **270** may be referred to as a pixel defining layer. For example, the bank layer **270** may be disposed to have a plurality of openings, and each opening may correspond to an emitting area EA of each emitting element **260**.
(78) The bank layer **270** may include an organic material such as acrylic resin, epoxy resin, phenolic resin, polyamide resin and polyimide resin. The embodiments of the present disclosure are not limited thereto. In one embodiment, the bank layer **270** includes black material such as a black matrix or black resin or dye. For example, the bank layer **270** may have a same material as the light absorbing pattern **317** further described below.
(79) The emitting layer **262** is disposed on the first electrode **261** and the bank layer **270**. The emitting layer **262** may include a hole transporting layer, at least one emitting material layer and an electron transporting layer. The embodiments of the present disclosure are not limited thereto. When a voltage is applied to the first electrode **261** and the second electrode **263**, a hole and an electron move to the emitting material layer through the hole transporting layer and the electron transporting layer, respectively, and the hole and the electron are combined in the emitting material layer to emit a light.
(80) The emitting layer **262** may have a white emitting layer emitting a white colored light. Alternatively, the emitting layer **262** may have a red emitting layer emitting a red colored light, a green emitting layer emitting a green colored light and a blue emitting layer emitting a blue colored light. The emitting layer **262** may be disposed to cover the first electrode **261** and the bank layer **270**.
(81) The second electrode **263** is disposed on the emitting layer **262**. When the display device **100** has a top emission type, the second electrode **263** may include a transparent conductive oxide (TCO) such as indium tin oxide (ITO) and indium zinc oxide (IZO) or a semi-transmissive conductive material such as magnesium (Mg), silver (Ag) and an alloy thereof to transmit a light. The embodiments of the present disclosure are not limited thereto.
(82) An encapsulating layer **280** is disposed on the plurality of emitting elements **260** to cover the plurality of emitting elements **260**. The encapsulating layer **280** extends from the display area DA to the non-display area NDA.
(83) The encapsulating layer **280** may prevent or at least reduce penetration of an oxygen or a moisture to the emitting layer **262** and the second electrode **263**. The encapsulating layer **280** may include at least one inorganic layer and at least one organic layer. For example, the encapsulating

layer **280** may include a first inorganic encapsulating layer **281**, an organic encapsulating layer **282** and a second inorganic encapsulating layer **283**.

(84) The first inorganic encapsulating layer **281** may be disposed on the second electrode **263**. The first inorganic encapsulating layer **281** may be formed to cover the second electrode **263**. The organic encapsulating layer **282** may be disposed on the first inorganic encapsulating layer **281**. The organic encapsulating layer **282** may have a predetermined thickness to prevent or at least reduce penetration of particles to the emitting layer **262** and the second electrode **263** through the first inorganic encapsulating layer **281**. The second inorganic encapsulating layer **283** may be disposed on the organic encapsulating layer **282**. The second inorganic encapsulating layer **283** may be formed to cover the organic encapsulating layer **282**.

(85) The first and second inorganic encapsulating layers **281** and **283** may include one of silicon nitride, aluminum nitride, zirconium nitride, titanium nitride, hafnium nitride, tantalum nitride, silicon oxide, aluminum oxide and titanium oxide. The embodiments of the present disclosure are not limited thereto.

(86) The organic encapsulating layer **282** may include one of acrylic resin and epoxy resin.

(87) The display device **100** may include the dam DAM having a closed curve shape in the non-display area NDA to surround the organic encapsulating layer **282**. The dam DAM may be disposed to surround an edge portion of the display area DA and may block a flow of the organic encapsulating layer **282**. As a result, the dam DAM may prevent that the organic encapsulating layer **282** is exposed outside the display device **100** or the organic encapsulating layer **282** intrudes the pad area PA.

(88) Although the dam DAM may include a single dam, the flow of the organic encapsulating layer **282** may be effectively blocked by a plurality of dams. Although two dams DAM are exemplarily disposed in FIG. **7**, it is not limited thereto.

(89) The dam DAM may be simultaneously formed with the first planarizing layer **250** or the bank layer **270** of the subpixel P and may include the same material as the first planarizing layer **250** or the bank layer **270** of the subpixel P. In one embodiment, the dam DAM may include organic material.

(90) The second inorganic encapsulating layer **283** may be disposed to cover the dam DAM.

(91) The touch sensing layer is disposed on the encapsulating layer **280**. The touch sensing layer may include a plurality of touch electrodes **320** having a plurality of first touch electrodes TE and a plurality of second touch electrodes RE, a plurality of bridge electrodes BE, a touch buffer layer **311** and an insulating layer **313**.

(92) The touch buffer layer **311** is disposed on the encapsulating layer **280** in the display area DA to expose the pad PAD in the non-display area NDA. The touch buffer layer **311** may be formed to cover the dam DAM.

(93) The touch buffer layer **311** may block or at least reduce penetration of a solution such as a developer or an etchant used in a fabrication process of the plurality of touch electrodes on the touch buffer layer **311** or a particle such as an external moisture to the emitting element of an organic material.

(94) The plurality of bridge electrodes BE are disposed on the touch buffer layer **311**. The plurality of bridge electrodes BE is disposed in the display area DA and electrically connect the plurality of first touch electrodes TE on the insulating layer **313**.

(95) To prevent an electric shortage of the plurality of first touch electrodes TE and the plurality of second touch electrodes RE at crossing thereof, as shown in FIG. **5**, the first touch electrodes TE adjacent to each other along a first direction (y axis direction) are electrically connected to each other through the bridge electrode BE. The plurality of bridge electrodes BE may be disposed in a different layer from the plurality of first touch electrodes TE and the plurality of second touch electrodes RE and may be connected to the adjacent first touch electrodes TE through contact holes CH. The plurality of bridge electrodes BE may cross the plurality of second touch electrodes RE.

(96) The contact holes CH may be formed in the insulating layer **313**. The plurality of bridge electrodes BE are disposed under the insulating layer **313** to be exposed through the two contact holes CH. As a result, one bridge electrode BE is connected to adjacent two first touch electrodes TE.
(97) The insulating layer **313** is disposed on the touch buffer layer **311** to cover the plurality of bridge electrodes BE. The insulating layer **313** may insulate the plurality of bridge electrodes BE and the plurality of second touch electrodes RE. The insulating layer **313** may be disposed between the plurality of bridge electrodes BE to insulate the plurality of bridge electrodes BE.
(98) The insulating layer **313** extends from the display area DA to the non-display area NDA. The insulating layer **313** may be formed to cover the dam DAM such that a step difference due to the dam DAM is reduced.
(99) The plurality of touch electrodes **320** of a mesh shape are disposed on the insulating layer **313**. The plurality of touch electrodes **320** includes the plurality of first touch electrodes TE and the plurality of second touch electrodes RE.
(100) The plurality of first touch electrodes TE and the plurality of second touch electrodes RE are disposed in the display area DA. The plurality of first touch electrodes TE are disposed along the first direction (y axis direction) to be connected to each other, and the plurality of second touch electrodes RE are disposed along a second direction (x axis direction) to be connected to each other. The first direction (y axis direction) may be a direction parallel to the plurality of scan lines S**1** to Sn, and the second direction (x axis direction) may be a direction parallel to the plurality of data lines D**1** to Dm. Alternatively, the first direction (y axis direction) may be a direction parallel to the plurality of data lines D**1** to Dm, and the second direction (x axis direction) may be a direction parallel to the plurality of scan lines S**1** to Sn.
(101) A line of the plurality of first touch electrodes TE connected along the first direction (y axis direction) is electrically insulated from an adjacent line of the plurality of first touch electrodes TE along the second direction (x axis direction). A line of the plurality of second touch electrodes RE connected along the second direction (x axis direction) is electrically insulated from an adjacent line of the plurality of second touch electrodes RE along the first direction (y axis direction).
(102) As a result, a mutual capacitance of the touch sensor may be formed at crossing of the first touch electrode TE and the second touch electrode RE.
(103) For example, as shown in FIG. **6**, the plurality of touch electrodes **320** may have a mesh shape having openings.
(104) Since the plurality of touch electrodes **320** have a mesh shape and the plurality of emitting elements **260** correspond to the openings of the plurality of touch electrodes **320**, a light extraction efficiency may be improved.
(105) The plurality of touch electrodes **320** may be disposed to correspond to the bank layer **270**. That is, the plurality of touch electrodes **320** overlap the bank layer **270**. The bank layer **270** has the plurality of openings and the plurality of openings of the bank layer **270** correspond to the emitting area EA. The openings of the plurality of touch electrodes **320** may be disposed to correspond to the plurality of openings of the bank layer **270**.
(106) Accordingly, the plurality of touch electrodes **320** may be disposed along the bank layer **270** to correspond to the bank layer **270**. Since the plurality of touch electrodes **320** are disposed to correspond to the bank layer **270** and the openings of the plurality of touch electrodes **320** are disposed to correspond to the emitting area EA, the openings of the plurality of touch electrodes **320** may overlap the emitting area EA and a reduction of an emission efficiency may be minimized.
(107) The pad PAD is disposed in the non-display area NDA, and a touch routing line **330** may be disposed on the insulating layer **313** to electrically connect the pad PAD and the plurality of touch electrodes **320**.
(108) The pad PAD may be formed of the same material and in the same layer as the gate electrode **212**, and the touch routing line **330** may be formed of the same material and in the same layer as

the plurality of touch electrodes **320**.

(109) In FIGS. **7** and **8**, a first barrier layer **312** may be disposed on the insulating layer **313** to correspond to a non-light emitting area NEA in the display area DA. The first barrier layer **312** may be a layer blocking a light of a viewing angle direction and focusing a light toward a front direction of the display panel. The front direction may be a direction perpendicular (e.g., normal) to the display surface of the display panel in one embodiment. As shown in FIGS. **7** and **8**, one or more intermediate layers (e.g., encapsulating layer **280**, touch buffer layer **311**, and insulating layer **313**) may be between the first barrier layer **312** and the light emitting element **260**. The first barrier layer **312** may include a metallic material or a material absorbing a light. The metallic material may be configured to reflect light. The first barrier layer **312** may include the same material as the plurality of touch electrodes **320**. In one embodiment, the first barrier layer **312** is a portion of the touch electrodes **320** that are disposed in the mesh shape. Thus, the touch electrodes **320** are configured to block light of a viewing angle direction and focus the light toward the front direction of the display panel since the touch electrodes **320** function as a barrier layer as well as sense touch of the display panel. The embodiments of the present disclosure are not limited thereto. Since the plurality of touch electrodes **320** are utilized as the first barrier layer **312**, a fabrication time and a fabrication cost may be reduced without an additional process. The first barrier layer **312** may block a light **8***f* and **8***g* (of FIG. **8**) of a viewing angle direction other than a front direction.

(110) As a result, the display device **100** according to the first embodiment of the present disclosure may control or cut off an up and down viewing angle without attaching a separate light control film on the display panel **110**. In one embodiment, the up and down viewing angle is a polar angle along the Y-axis direction with respect to the front direction.

(111) A first protecting layer **350** covering the plurality of touch electrodes **320** and the touch routing line **330** may be disposed on the first barrier layer **312**.

(112) A plurality of lenses **333** corresponding to the plurality of emitting areas EA, respectively, may be disposed on the first protecting layer **350**. That is, each lens **333** overlaps a corresponding emission area EA from the plurality of emission areas EA. A second protecting layer **360** may be disposed on the plurality of lenses **333**. The first protecting layer **350** and the second protecting layer **360** may be formed to expose the pad PAD in the non-display area NDA.

(113) The plurality of lenses **333** may be disposed on the plurality of emitting elements **260** so as to overlap the plurality of emitting elements **260**. As shown in FIGS. **7** and **8**, the first barrier layers **312** are closer to the substrate **111** than the plurality of lenses **333**. The plurality of lenses **333** may be disposed on first barrier layer **312** and overlap the emitting area EA. Each lens **333** may extend along the first direction (y axis direction) to have a size equal to or greater than each emitting element **260**. The plurality of lenses **333** may be disposed parallel to each other along the first direction (y axis direction). For example, the first direction (y axis direction) may be a short axis direction, a vertical direction or an up and down direction in a display device of rectangular shape of 16:9.

(114) In FIG. **8**, the plurality of lenses **333***a*, **333***b* and **333***c* configured to refract light are disposed to correspond to the plurality of emitting areas EA of the plurality of subpixels P**1**, P**2** and P**3**.

(115) The plurality of lenses **333***a*, **333***b* and **333***c* may be disposed to correspond to the plurality of subpixels P**1**, P**2** and P**3**. For example, a first lens **333***a* is disposed to correspond to the emitting area EA of a first subpixel P**1**, a second lens **333***b* is disposed to correspond to the emitting area EA of a second subpixel P**2**, and a third lens **333***c* is disposed to correspond to the emitting area EA of a third subpixel P**3**. That is, the first lens **333***a* overlaps the first subpixel P**1**, the second lens **333***b* overlaps the second subpixel P**2**, and third lens **333***c* overlaps the third subpixel P**3**.

(116) For example, the plurality of lenses **333***a*, **333***b* and **333***c* may be a convex lens having a semispherical shape convex toward a polarizing plate **500** along a third direction (z axis direction).

(117) FIG. **9** is a magnified cross-sectional view showing a lens of FIG. **7** according to the first embodiment.

(118) In FIG. **9**, when a light is incident to the plurality of lenses **333***a*, **333***b* and **333***c* from the emitting layers **241**, **242** and **243**, the light is refracted to have a refraction angle θ2 greater than an incidence angle θ1 (θ1<θ2). Since an incidence angle of the light passing through the plurality of lenses **333***a*, **333***b* and **333***c* to the second substrate **112** is reduced, the light may not be totally reflected. As a result, the display device **100** according to a first embodiment of the present disclosure may have an excellent light extraction efficiency.
(119) The plurality of lenses **333***a*, **333***b* and **333***c* may focus a light **8***a* and **8***b* of a front direction of the display device **100** among the light emitted from the emitting element **260**.
(120) A refractive index of the plurality of lenses **333** may be greater than a refractive index of the second protecting layer **360**. Since the refractive index of the second protecting layer **360** is less than the refractive index of the plurality of lenses **333**, a focusing efficiency of a light of a front direction and a light extraction efficiency may be improved.
(121) Since the refractive index of the plurality of lenses **333** is greater than the refractive index of the second protecting layer **360**, a focusing effect that the light incident to each lens **333** is refracted toward a thick portion of each lens **333** is obtained and a luminance of a front direction is improved. Since the light is transmitted from a medium of a relatively high refractive index to a medium of a relatively low refractive index, the light is not leaked and the luminance of a front direction increases.
(122) For example, the lens **333** may have a refractive index in a range of 1.5 to 1.8, and the second protecting layer **360** may have a refractive index in a range of 1.3 to 1.5. As a refractive index difference increases, a focusing efficiency of a light of a front direction increases.
(123) The lens **333** may be disposed to include the emitting area EA of the emitting element **260**. When the lens **333** has a size smaller than the emitting area EA, an amount of a light passing through the lens **333** decreases, and the focusing efficiency of the lens **333** and the light extraction efficiency are reduced.
(124) When the lens **333** has a size greater than the emitting area EA, a curvature radius R of the lens **333** increases, and a power of the lens **333** decreases. As a result, the focusing efficiency of the lens **333** is reduced.
(125) As a result, the lens **333** may have a size such that a power of the lens **333** increases by reducing the curvature radius R and a focusing efficiency is maximized by making the light emitted from the emitting area EA pass through the lens **333**.
(126) Although the lens **333** may include photoacryl (PAC) which can be formed through a relative low temperature process lower than about 100 degrees, it is not limited thereto. For example, the lens **333** may include polytriazine or polytriazine having at least one of titanium oxide (TiO_2), zirconium oxide (ZrO_2) and nano-filler.
(127) The lens **333** may be formed to have a convex shape through an exposure process.
(128) The emitting element **260** is disposed under the lens **333** so as to overlap the lens **333**. When the lens **333** is formed through a relatively high temperature process, the emitting element **260** may be damaged during the relatively high temperature process. As a result, the lens **333** includes a material which can be formed through a relatively low temperature process lower than about 100 degrees, and the damage of the emitting element **260** during the process of forming the lens **333** is reduced.
(129) The lens **333** may be formed through the following process. A lens layer is coated on the first protecting layer **350**. The first protecting layer **350** may include photoacryl. The lens layer may include polytriazine or polytriazine having at least one of titanium oxide (TiO_2), zirconium oxide (ZrO_2) and nano-filler for a relatively low temperature process.
(130) After the lens layer is coated, an edge bead removal process of removing an edge portion is performed. Next, a pre-bake process of removing a solvent in the lens layer is performed. Next, an exposure process in which the lens layer is exposed to a light through a mask to cause a photoreaction is performed.

(131) Next, a develop process, a rinse process and a cure process are performed to form the lens **333**.
(132) The second protecting layer **360** is disposed on the lens **333** to cover the lens **333**. Since the refractive index of the second protecting layer **360** is less than the refractive index of the lens **333**, a focusing efficiency of a light and a light extraction efficiency may further increase due to a refractive index difference at an interface of the lens **333**.
(133) The display device **100** according to a first embodiment of the present disclosure may further include the polarizing plate **500** on the second protecting layer **360**. A reflection of an external light may be reduced due to the polarizing plate **500**.
(134) FIG. **10** is a graph showing a normalized transmittance intensity with respect to an up and down viewing angle of a display device according to the first embodiment of the present disclosure.
(135) In FIG. **10**, a display device according to a comparison example (**10***a*) does not include a lens **333** and a first barrier layer **312** and includes a light control film on a display panel. A display device according to a first embodiment (**10***b*) includes a lens **333** and a first barrier layer **312** and does not include a light control film.
(136) The up and down viewing angle of the display device according to a first embodiment (**10***b*) is less than the up and down viewing angle of the display device according to a comparison example (**10***a*). Although the display device according to a comparison example (**10***a*) has the up and down viewing angle (e.g., a viewing angle range) of about ±35 degrees, the display device according to a first embodiment (**10***b*) has the up and down viewing angle of about ±20 degrees. As a result, the up and down viewing angle of the display device is improved.
(137) FIG. **11** is a cross-sectional view showing a display device according to a second embodiment of the present disclosure, and FIG. **12** is a magnified cross-sectional view of a region **11**A of FIG. **11** according to the second embodiment of the present disclosure. FIG. **11** corresponds to lines I-I′ and II-II′ of FIGS. **4** and **5**.
(138) Illustration on parts of the second embodiment the same as those of the first embodiment may be omitted, and the parts of the first embodiment may be intactly applied to those of the second embodiment.
(139) In FIGS. **11** and **12**, a second barrier layer **315** may be disposed on the encapsulating layer **280** or on the touch buffer layer **311**. The second barrier layer **315** may be a layer blocking a light of a viewing angle direction and focusing a light of a front direction. The second barrier layer **315** may include a metallic material or a material absorbing a light. The metallic material may be reflective. The insulating layer **313** may be disposed on the second barrier layer **315**. The first barrier layer **312** and the lens **343** are disposed on the insulating layer **313**. The first barrier layer **312** may be disposed between the plurality of lenses **343***a*, **343***b* and **343***c*. For example, the plurality of lenses **343***a*, **343***b* and **343***c* may be disposed to correspond to the emitting area EA. The plurality of lenses **343***a*, **343***b* and **343***c* may be disposed to be non-overlapping with the second barrier layer **315** and overlap the emitting area EA. The first barrier layer **312** and the second barrier layer **315** may be disposed in the non-light emitting area NEA that is disposed between a pair of emission areas EA. As shown in FIGS. **11** and **12**, a first barrier layer **312** is between a pair of lenses **343**. The first barrier layer **312** and the second barrier layer **315** may include a metallic material. In one embodiment, the first barrier layer **312** and the second barrier layer **315** include different materials. The first barrier layer **312** may include the same material as the plurality of touch electrodes **320** as the first barrier layer **312** is a portion of the touch electrodes **320** (e.g., first touch electrodes) that are disposed in the mesh shape as previously described above with respect to FIGS. **7** and **8**. The embodiments of the present disclosure are not limited thereto. The second barrier layer **315** may include the same material as the plurality of bridge electrodes BE. In one embodiment, the second barrier layer **315** is a portion of the touch electrodes **320** (e.g., second touch electrodes) that are disposed in the mesh shape as previously described above with respect to FIGS. **7** and **8**. The embodiments of the present disclosure are not limited thereto. The

insulating layer **313** may be disposed on the second barrier layer **315**. The first barrier layer **312** may be disposed on the insulating layer **313** in the non-light emitting area NEA. A fabrication time and a fabrication cost may be reduced without an additional process since the first barrier layer **312** is the touch electrodes **320** and the second barrier layer **315** is made of a same material as the bridge electrodes BE. The second barrier layer **315** may block a light **8***f* and **8***g* (of FIG. **8**) of a viewing angle direction. Since the first barrier layer **312** and the second barrier layer **315** are disposed to correspond to the non-light emitting area NEA, a leakage light **12***c* between the lenses **343** may be blocked.

(140) Accordingly, the display device according to the second embodiment of the present disclosure may control or cut off an up and down viewing angle without attaching a separate light control film on the display panel.

(141) A protecting layer **350** may be disposed on the first barrier layer **312**, the plurality of lenses **343**, the plurality of touch electrodes **320** and the touch routing line **330**.

(142) The plurality of lenses **343***a*, **343***b* and **343***c* may be disposed on the insulating layer **313** to correspond to the plurality of emitting areas EA, respectively. The protecting layer **350** may be disposed on the plurality of lenses **343**. The protecting layer **350** may be formed to expose the pad PAD in the non-display area NDA.

(143) The plurality of lenses **343** may be disposed on the plurality of emitting elements **260**. Each lens **343** may extend along the first direction (y axis direction) to have a size equal to or greater than each emitting element **260**. The plurality of lenses **343** may be disposed parallel to each other along the first direction (y axis direction). For example, the first direction (y axis direction) may be a short axis direction, a vertical direction or an up and down direction in a display device of rectangular shape of 16:9.

(144) Since the focusing of a light of the plurality of lenses **343** of a second embodiment is the same as that of a first embodiment, illustration on the focusing of a light of the plurality of lenses **343** may be omitted.

(145) The display device according to a second embodiment of the present disclosure may further include the polarizing plate **500** on the protecting layer **350**. A reflection of an external light may be reduced due to the polarizing plate **500**.

(146) FIG. **13** is a graph showing a normalized transmittance intensity with respect to an up and down viewing angle of a display device according to the second embodiment of the present disclosure.

(147) In FIG. **13**, a display device according to a comparison example (**13***a*) does not include a lens **343**, a first barrier layer **312** and a second barrier layer **315** and includes a light control film on a display panel. A display device according to a second embodiment (**13***b*) includes a lens **343**, a first barrier layer **312** and a second barrier layer **315** and does not include a light control film.

(148) The up and down viewing angle of the display device according to the second embodiment (**13***b*) is less than the up and down viewing angle of the display device according to a comparison example (**13***a*). Although the display device according to a comparison example (**13***a*) has the up and down viewing angle of about ±35 degrees, the display device according to a second embodiment (**13***b*) has the up and down viewing angle of about ±30 degrees. As a result, the up and down viewing angle of the display device is improved.

(149) FIG. **14** is a cross-sectional view showing a display device according to a third embodiment of the present disclosure, and FIG. **15** is a magnified cross-sectional view of a region **14**A of FIG. **14** according to the third embodiment of the present disclosure. FIG. **14** corresponds to lines I-I′ and II-II′ of FIGS. **4** and **5**.

(150) Illustration on parts of the third embodiment the same as those of the first embodiment may be omitted, and the parts of the first embodiment may be intactly applied to those of the third embodiment.

(151) In FIGS. **14** and **15**, a light absorbing pattern **317** may be disposed on the encapsulating layer

280 or on the touch buffer layer **311**. The light absorbing pattern **317** may be a black matrix of black resin or dye. The embodiments of the present disclosure are not limited thereto. The light absorbing pattern **317** may be referred to as a second barrier layer. The insulating layer **313** may be disposed on the light absorbing pattern **317**. The first barrier layer **312** and the lens **343** may be disposed on the insulating layer **313**. The first barrier layer **312** may be disposed between the plurality of lenses **343***a*, **343***b* and **343***c*. For example, the plurality of lenses **343***a*, **343***b* and **343***c* may be disposed to correspond to (e.g., overlap) the emitting area EA. The first barrier layer **312** and the light absorbing pattern **317** may be disposed in the non-light emitting area NEA. The first barrier layer **312** may include the same material as the plurality of touch electrodes **320**. The embodiments of the present disclosure are not limited thereto. Since the light absorbing pattern **317** is disposed to correspond to the non-light emitting area NEA, a leakage light **12***c* between the lenses **343** may be blocked. When the light absorbing pattern **317** is not disposed, a first reflection light **12***d* (of FIG. **12**) reflected on the second barrier layer **315** and a second reflection light **12***e* (of FIG. **12**) reflected on the source electrode **213** and the drain electrode **214** may pass through the plurality of lenses **343**. When the light absorbing pattern **317** is disposed to correspond to the non-light emitting area NEA, the first reflection light **15***e* and the second reflection light **15***d* may be blocked by the light absorbing pattern **317**.
(152) As a result, when the light absorbing pattern **317** of a black material is disposed to correspond to the non-light emitting area NEA, reflection on the inner metal layer may be reduced and a leakage light through the lenses **343** may be prevented.
(153) Although not shown, only the light absorbing pattern **317** may be disposed on the touch buffer layer **311** and the first barrier layer **312** may not be disposed on the touch buffer layer **311**. The embodiments of the present disclosure are not limited thereto.
(154) A protecting layer **350** may be disposed on the first barrier layer **312**, the plurality of lenses **343**, the plurality of touch electrodes **320** and the touch routing line **330**.
(155) The plurality of lenses **343***a*, **343***b* and **343***c* may be disposed on the insulating layer **313** to correspond to the plurality of emitting areas EA, respectively. The protecting layer **350** may be disposed on the plurality of lenses **343**. The protecting layer **350** may be formed to expose the pad PAD in the non-display area NDA.
(156) The plurality of lenses **343** may be disposed on the plurality of emitting elements **260**. Each lens **343** may extend along the first direction (y axis direction) to have a size equal to or greater than each emitting element **260**. The plurality of lenses **343** may be disposed parallel to each other along the first direction (y axis direction). For example, the first direction (y axis direction) may be a short axis direction, a vertical direction or an up and down direction in a display device of rectangular shape of 16:9.
(157) Since the focusing of a light of the plurality of lenses **343** of a third embodiment is the same as that of a first embodiment, illustration on the focusing of a light of the plurality of lenses **343** may be omitted.
(158) The display device according to a third embodiment of the present disclosure may further include the polarizing plate **500** on the protecting layer **350**. A reflection of an external light may be reduced due to the polarizing plate **500**.
(159) FIG. **16** is a graph showing a normalized transmittance intensity with respect to an up and down viewing angle of a display device according to a third embodiment of the present disclosure.
(160) In FIG. **16**, a display device according to a comparison example (**16***a*) does not include a lens **343**, a light absorbing pattern **317** and a first barrier layer **312** and includes a light control film on a display panel. A display device according to a third embodiment (**16***b*) includes a lens **343**, a light absorbing pattern **317** and a first barrier layer **312** and does not include a light control film.
(161) The up and down viewing angle of the display device according to a third embodiment (**16***b*) is smaller than the up and down viewing angle of the display device according to a comparison example (**16***a*). Although the display device according to a comparison example (**16***a*) has the up

and down viewing angle of about ±35 degrees, the display device according to a third embodiment (**16***b*) has the up and down viewing angle of about ±30 degrees. As a result, the up and down viewing angle of the display device is improved. In FIG. **12**, an inversion of a viewing angle luminance **8***p* may occur due to an additional leakage light, for example, the second reflection light **8***e*. In FIG. **16**, an inversion of a viewing angle luminance may not occur due to the light absorbing pattern **317**.
(162) FIG. **17** is a cross-sectional view showing a display device according to a fourth embodiment of the present disclosure. FIG. **17** corresponds to a region **14**A of FIG. **14**.
(163) Illustration on parts of the fourth embodiment the same as those of the first embodiment may be omitted, and the parts of the first embodiment may be intactly applied to those of the fourth embodiment.
(164) In FIG. **17**, a light absorbing pattern **317** may be disposed on the encapsulating layer **280** or on the touch buffer layer **311**. The light absorbing pattern **317** may be a black matrix of black resin or dye. The embodiments of the present disclosure are not limited thereto. The insulating layer **313** may be disposed on the light absorbing pattern **317**. The first barrier layer **312** and the lens **343** may be disposed on the insulating layer **313**. The first barrier layer **312** may be disposed between the plurality of lenses **343***a*, **343***b* and **343***c*. For example, the plurality of lenses **343***a*, **343***b* and **343***c* may be disposed to correspond to (e.g., overlap) the emitting area EA. The first barrier layer **312** and the light absorbing pattern **317** may be disposed to correspond to the non-light emitting area NEA. The first barrier layer **312** may include the same material as the plurality of touch electrodes **320**. The embodiments of the present disclosure are not limited thereto. Since the light absorbing pattern **317** is disposed to correspond to the non-light emitting area NEA, a leakage light **12***c* (of FIG. **12**) between the lenses **343** may be blocked. When the light absorbing pattern **317** is not disposed, a first reflection light **12***d* (of FIG. **12**) reflected on the second barrier layer **315** and a second reflection light **12***e* (of FIG. **12**) reflected on the source electrode **213** and the drain electrode **214** may pass through the plurality of lenses **343**. When the light absorbing pattern **317** is disposed to correspond to the non-light emitting area NEA, the first reflection light **15***e* and the second reflection light **15***d* may be blocked by the light absorbing pattern **317**.
(165) As a result, when the light absorbing pattern **317** of a black material is disposed to correspond to the non-emitting area NEA, reflection on the inner metal layer may be reduced and a leakage light through the lenses **343** may be prevented.
(166) The light absorbing pattern **317** may have a first width W**1**, and the first barrier layer **312** may have a second width W**2** that is smaller than the first width W**1**. Since the first width W**1** of the light absorbing pattern **317** is greater than the second width W**2** of the first barrier layer **312**, a light passing outside the light absorbing pattern **317** and reflected on the first barrier layer **312** may be reduced or minimized. The lens **343** may have a third width W**3**, and the emitting area EA may have a fourth width W**4** different from the third width W**3**. For example, the lens **343** may have a third width W**3** that is greater than the fourth width W**4**. Since the third width W**3** of the lens **343** is greater than the fourth width W**4** of the emitting area EA, a light **8***a* and **8***b* of a front direction through the lens **343** may increase or be maximized.
(167) As a result, when the light absorbing pattern **317** is wider than the first barrier layer **312**, reflection on the first barrier layer **312** may be reduced and a leakage light due to the first barrier layer **312** may be prevented. Further, when the lens **343** is wider than the emitting area EA, a light through the lens **343** may increase.
(168) Although not shown, only the light absorbing pattern **317** may be disposed on the touch buffer layer **311** and the first barrier layer **312** may not be disposed on the touch buffer layer **311**. The embodiments of the present disclosure are not limited thereto.
(169) A protecting layer **350** may be disposed on the first barrier layer **312**, the plurality of lenses **343**, the plurality of touch electrodes **320** and the touch routing line **330**.
(170) The plurality of lenses **343***a*, **343***b* and **343***c* may be disposed on the insulating layer **313** to

correspond to the plurality of emitting areas EA, respectively. The protecting layer **350** may be disposed on the plurality of lenses **343**. The protecting layer **350** may be formed to expose the pad PAD in the non-display area NDA.
(171) The plurality of lenses **343** may be disposed on the plurality of emitting elements **260**. Each lens **343** may extend along the first direction (y axis direction) to have a size equal to or greater than each emitting element **260**. The plurality of lenses **343** may be disposed parallel to each other along the first direction (y axis direction). For example, the first direction (y axis direction) may be a short axis direction, a vertical direction or an up and down direction in a display device of rectangular shape of 16:9.
(172) Since the focusing of a light of the plurality of lenses **343** of a fourth embodiment is the same as that of a first embodiment, illustration on the focusing of a light of the plurality of lenses **343** may be omitted.
(173) The display device according to a fourth embodiment of the present disclosure may further include the polarizing plate **500** on the protecting layer **350**. A reflection of an external light may be reduced due to the polarizing plate **500**.
(174) FIG. **18** is a cross-sectional view showing a display device according to a fifth embodiment of the present disclosure. FIG. **18** corresponds to a region **14**A of FIG. **14**. Illustration on parts of the fifth embodiment the same as those of the first embodiment may be omitted, and the parts of the first embodiment may be intactly applied to those of the fifth embodiment.
(175) In FIG. **18**, a light absorbing pattern **317** may be disposed on the encapsulating layer **280** or on the touch buffer layer **311**. The light absorbing pattern **317** may be a black matrix of black resin or dye. However, the embodiments of the present disclosure are not limited thereto. The insulating layer **313** may be disposed on the light absorbing pattern **317** as shown in FIG. **18**. The first barrier layer **312** and the lens **343** may be disposed on the insulating layer **313**. The first barrier layers **312** may be disposed between the plurality of lenses **343***a*, **343***b* and **343***c*. For example, the plurality of lenses **343***a*, **343***b* and **343***c* may be disposed to respectively correspond to (e.g., overlap) the emitting area EA of first, second and third subpixels P**1**, P**2** and P**3**. Each of the first barrier layers **312** and the light absorbing patterns **317** may be disposed to correspond to (e.g., overlap) the non-light emitting area NEA of a corresponding one of the first, second, and third subpixels P**1**, P**2** and P**3**. The first barrier layer **312** may include the same material as the plurality of touch electrodes **320**. However, the embodiments of the present disclosure are not limited thereto.
(176) Since the light absorbing pattern **317** is disposed to correspond to the non-light emitting area NEA, a leakage light **12***c* (of FIG. **12**) between the lenses **343** may be blocked. When the light absorbing pattern **317** is not disposed, a first reflection light **12***d* (of FIG. **12**) reflected on the second barrier layer **315** and a second reflection light **12***e* (of FIG. **12**) reflected on the source electrode **213** and the drain electrode **214** may pass through the plurality of lenses **343**. When the light absorbing pattern **317** is disposed to correspond to the non-light emitting area NEA, the first reflection light **15***e* and the second reflection light **15***d* may be blocked by the light absorbing pattern **317**.
(177) As a result, when the light absorbing pattern **317** of a black material is disposed to correspond to the non-emitting area EA, reflection on the inner metal layer may be reduced and a leakage light through the lenses **343** may be prevented.
(178) The light absorbing patterns **317** of the first, second and third subpixels P**1**, P**2** and P**3** may have different widths. For example, the light absorbing patterns **317** of the first, second and third subpixels P**1**, P**2** and P**3** may have first-first, first-second and first-third widths W**11**, W**12** and W**13**, respectively where the widths are different from each other. The first barrier layers **312** of the first, second and third subpixels P**1**, P**2** and P**3** may have different widths. For example, the first barrier layers **312** of the first, second and third subpixels P**1**, P**2** and P**3** have second-first, second-second, second-third widths W**21**, W**22** and W**23**, respectively, where the widths are different from each other. The first-first width W**11** may be smaller than the second-first width W**21**, the first-

second width W**12** may be smaller than the second-second width W**22**, and the first-third width W**13** may be smaller than the second-third width W**23**. As a result, a light passing outside the light absorbing pattern **317** and reflected on the first barrier layer **312** may be reduced or minimized.
(179) The lenses **343***a*, **343***b* and **343***c* of the first, second and third subpixels P**1**, P**2** and P**3** may have different widths. For example, the lenses **343***a*, **343***b* and **343***c* of the first, second and third subpixels P**1**, P**2** and P**3** may have third-first, third-second and third-third widths W**31**, W**32** and W**33**, respectively, where the widths are different from each other. The emitting areas EA of the first, second and third subpixels P**1**, P**2** and P**3** may have different widths. For example, the emitting areas EA of the first, second and third subpixels P**1**, P**2** and P**3** have fourth-first, fourth-second and fourth-third widths W**41**, W**42** and W**43**, respectively, where the widths are different from each other. The third-first width W**31** may be greater than the fourth-first width W**41**, the third-second width W**32** may be greater than the fourth-second width W**42**, and the third-third width W**33** may be greater than the fourth-third width W**43**. As a result, a light **8***a* and **8***b* of a front direction through the lens **343** may increase or be maximized.
(180) For example, the fourth-first width W**41** corresponding to the first subpixel P**1** of a red color may be smaller than the fourth-second width W**42** corresponding to the second subpixel P**2** of a green color and may be greater than the fourth-third width W**43** corresponding to the third subpixel P**3** of a blue color based on a luminance and a lifetime. Similarly, the third-first width W**31** of the lens **343***a* corresponding to the first subpixel P**1** of the red color may be smaller than the third-second width W**32** of the lens **343***b* corresponding to the second subpixel P**2** of the green color and may be greater than the third-third width W**33** of the lens **343***c* corresponding to the third subpixel P**3** of the blue color.
(181) When the light absorbing pattern **317** is wider than the first barrier layer **312**, reflection on the first barrier layer **312** may be reduced and a leakage light due to the first barrier layer **312** may be prevented. Further, when the lens **343** is wider than the emitting area EA, a light through the lens **343** may increase.
(182) Although not shown, only the light absorbing pattern **317** may be disposed on the touch buffer layer **311** and the first barrier layer **312** may not be disposed on the touch buffer layer **311** in one embodiment. The embodiments of the present disclosure are not limited thereto.
(183) A protecting layer **350** may be disposed on the first barrier layer **312**, the plurality of lenses **343**, the plurality of touch electrodes **320** and the touch routing line **330**.
(184) The plurality of lenses **343***a*, **343***b* and **343***c* may be disposed on the insulating layer **313** to correspond to the plurality of emitting areas EA, respectively. The protecting layer **350** may be disposed on the plurality of lenses **343**. The protecting layer **350** may be formed to expose the pad PAD in the non-display area NDA.
(185) The plurality of lenses **343** may be disposed on the plurality of emitting elements **260**. Each lens **343** may extend along the first direction (y axis direction) to have a size equal to or greater than each emitting element **260**. The plurality of lenses **343** may be disposed parallel to each other along the first direction (y axis direction). For example, the first direction (y axis direction) may be a short axis direction, a vertical direction or an up and down direction in a display device of rectangular shape of 16:9.
(186) Since the focusing of a light of the plurality of lenses **343** of the fifth embodiment is the same as that of a first embodiment, illustration on the focusing of a light of the plurality of lenses **343** may be omitted.
(187) The display device according to a fifth embodiment of the present disclosure may further include the polarizing plate **500** on the protecting layer **350**. A reflection of an external light may be reduced due to the polarizing plate **500**.
(188) Consequently, in the display device according to an embodiment of the present disclosure, since a light control layer including at least one lens and a barrier layer is disposed on an encapsulating layer covering an emitting element, an up and down viewing angle and/or a right and

left viewing angle may be controlled or cut off and a front luminance may be improved due to increase of a light extraction efficiency.
(189) Further, a power consumption is reduced due to increase of a front luminance.
(190) Example embodiments of the present disclosure can also be described as follows:
(191) According to an example embodiment of a present disclosure, a display device may include: a display panel including a display area and a non-display area, the display area having a light emitting area and a non-light emitting area; at least one first barrier layer on the non-light emitting area of the display panel, the at least one first barrier layer configured to block light emitted from the light emitting area; at least one second barrier layer overlapping the at least one first barrier layer in the non-display area, the at least one second barrier layer configured to block light emitted from the light emitting area; and at least one lens overlapping the light emitting area and on a same layer as the at least one first barrier layer.
(192) In some example embodiments, the at least one first barrier layer may include at least a portion of a first touch electrode comprising a mesh shape.
(193) In some example embodiments, the at least one second barrier layer may include a black material configured to absorb the light.
(194) In some example embodiments, a width of the at least one first barrier layer may be different from a width of the at least one second barrier layer.
(195) In some example embodiments, the width of the at least one second barrier layer may be greater than the width of the at least one first barrier layer.
(196) In some example embodiments, a width of the at least one lens may be different from a width of the light emitting area.
(197) In some example embodiments, the width of the at least one lens may be greater than the width of the light emitting area.
(198) In some example embodiments, the display area may include a first subpixel, a second subpixel, and a third subpixel each having a corresponding light emitting area and a corresponding non-light emitting area, and the at least one lens includes a plurality of lens including a first lens corresponding to the first subpixel, a second lens corresponding to the second subpixel, and a third lens corresponding to the third subpixel, wherein a width of the first lens corresponding to the first subpixel, a width of the second lens corresponding to the second subpixel, and a width of the third lens corresponding to the third subpixel are different from each other, and wherein the at least one first barrier layer includes a plurality of first barrier layers and a width of a first barrier layer from the plurality of first barrier layers that corresponds to the first subpixel, a width of a first barrier layer from the plurality of first barrier layers that corresponds to the second subpixel, and a width of a first barrier layer from the plurality of first barrier layers that corresponds to the third subpixel are different from each other.
(199) In some example embodiments, a width of the first lens corresponding to the first subpixel may be greater than a width of the light emitting area corresponding to the first subpixel, a width of the second lens corresponding to the second subpixel may be greater than a width of the light emitting area corresponding to the second subpixel, and a width of the third lens corresponding to the third subpixel may be greater than a width of the light emitting area corresponding to the third subpixel.
(200) In some example embodiments, the display device may further include: an insulating layer between the at least one first barrier layer and the at least one second barrier layer.
(201) In some example embodiments, the at least one second barrier layer may include at least a portion of a second touch electrode comprising a mesh shape.
(202) In some example embodiments, the display device may further include: a protecting layer covering the at least one lens, and a refractive index of the at least one lens may be greater than a refractive index of the protecting layer.
(203) In some example embodiments, the at least one first barrier layer may be between a pair of

lenses including the at least one lens.
(204) In some example embodiments, the at least one first barrier layer and the at least one second barrier layer may include different materials.
(205) According to another example embodiment of the present disclosure, a display device may include: a substrate including a display area and a non-display area; a plurality of emitting elements in the display area, the plurality of emitting elements configured to emit light; at least one encapsulating layer on the plurality of emitting elements; at least one barrier layer on the at least one encapsulating layer and non-overlapping with an emitting element from the plurality of emitting elements, the at least one barrier layer configured to block the light emitted by the plurality of emitting elements; and a plurality of lenses on the at least one encapsulating layer, the plurality of lenses overlapping the plurality of emitting elements.
(206) In some example embodiments, the plurality of lenses may be non-overlapping with the at least one barrier layer.
(207) In some example embodiments, the at least one barrier layer may include at least a portion of a touch electrode comprising a mesh shape.
(208) In some example embodiments, the at least one barrier layer may include a black material configured to absorb the light.
(209) In some example embodiments, the display device may further include: a protecting layer on the plurality of lenses, and a refractive index of the plurality of lenses may be greater than a refractive index of the protecting layer.
(210) In some example embodiments, the at least one barrier layer may include: a first barrier layer in a same layer as the plurality of lenses; and a second barrier layer that is between the first barrier layer and the substrate, the second barrier layer overlapping the first barrier layer.
(211) In some example embodiments, the display device may further include: a bridge electrode connected to the touch electrode, and the second barrier layer may include a same material as the bridge electrode.
(212) In some example embodiments, the second barrier layer may include a black material configured to absorb the light.
(213) According to another example embodiment of the present disclosure, a display device may include: a substrate including a display area and a non-display area; a plurality of light emitting elements in the display area, the plurality of light emitting elements configured to emit light; one or more intermediate layers on the plurality of light emitting elements; a plurality of touch electrodes on the one or more intermediate layers and non-overlapping with the plurality of light emitting elements, the plurality of touch electrodes configured to sense touch and block the light emitted from the plurality of light emitting elements; and a plurality of lenses that are overlapping with the plurality of light emitting elements.
(214) In some example embodiments, the plurality of lenses may be non-overlapping with the plurality of touch electrodes.
(215) In some example embodiments, the plurality of touch electrodes may be closer to the substrate than the plurality of lenses.
(216) In some example embodiments, the plurality of touch electrodes may be on a same layer as the plurality of lenses.
(217) In some example embodiments, at least one touch electrode from the plurality of touch electrodes may be between a pair of lenses from the plurality of lenses.
(218) In some example embodiments, the display device may further include: a plurality of barrier layers between the plurality of touch electrodes and the substrate, the plurality of barrier layers overlapping the plurality of touch electrodes and non-overlapping with the plurality of light emitting elements.
(219) In some example embodiments, the plurality of barrier layers may be non-overlapping with the plurality of lenses.

(220) In some example embodiments, the plurality of barrier layers may include black material configured to absorb the light.
(221) In some example embodiments, the plurality of barrier layers may include a reflective material.
(222) In some example embodiments, the display device may further include: a protecting layer covering the plurality of lenses, the protecting layer having a refractive index that is less than a refractive index of the plurality of lenses.
(223) It will be apparent to those skilled in the art that various modifications and variation can be made in the present disclosure without departing from the scope of the disclosure. Thus, it is intended that the present disclosure cover the modifications and variations of this disclosure, provided they come within the scope of the appended claims and their equivalents.

Claims

1. A display device, comprising: a display panel including a display area and a non-display area, the display area having a light emitting area and a non-light emitting area, and the non-display area having a pad area where pads are disposed; at least one first barrier layer on the non-light emitting area of the display panel, the at least one first barrier layer configured to block light emitted from the light emitting area; at least one second barrier layer overlapping the at least one first barrier layer in the non-light emitting area, the at least one second barrier layer configured to block light emitted from the light emitting area; at least one lens overlapping the light emitting area and on a same layer as the at least one first barrier layer without being on a layer at which the at least one second barrier layer is disposed; a touch buffer layer under the at least one second barrier layer; an insulating layer between the at least one first barrier layer and the at least one second barrier layer; and a protecting layer on the at least one lens, wherein at least one of the touch buffer layer, the insulating layer and the protecting layer extends to the pad area.
2. The display device of claim 1, wherein the at least one first barrier layer includes at least a portion of a first touch electrode comprising a mesh shape.
3. The display device of claim 2, wherein the at least one first barrier layer has a same material and a same layer as the first touch electrode.
4. The display device of claim 1, wherein the at least one second barrier layer includes a black material configured to absorb the light.
5. The display device of claim 1, wherein a width of the at least one first barrier layer is different from a width of the at least one second barrier layer.
6. The display device of claim 5, wherein the width of the at least one second barrier layer is greater than the width of the at least one first barrier layer.
7. The display device of claim 1, wherein a width of the at least one lens is different from a width of the light emitting area.
8. The display device of claim 7, wherein the width of the at least one lens is greater than the width of the light emitting area.
9. The display device of claim 1, wherein the display area includes a first subpixel, a second subpixel, and a third subpixel each having a corresponding light emitting area and a corresponding non-light emitting area, and the at least one lens includes a plurality of lens including a first lens corresponding to the first subpixel, a second lens corresponding to the second subpixel, and a third lens corresponding to the third subpixel, wherein a width of the first lens corresponding to the first subpixel, a width of the second lens corresponding to the second subpixel, and a width of the third lens corresponding to the third subpixel are different from each other, and wherein the at least one first barrier layer includes a plurality of first barrier layers and a width of a first barrier layer from the plurality of first barrier layers that corresponds to the first subpixel, a width of a first barrier layer from the plurality of first barrier layers that corresponds to the second subpixel, and a width

of a first barrier layer from the plurality of first barrier layers that corresponds to the third subpixel are different from each other.
10. The display device of claim 9, wherein a width of the first lens corresponding to the first subpixel is greater than a width of the light emitting area corresponding to the first subpixel, a width of the second lens corresponding to the second subpixel is greater than a width of the light emitting area corresponding to the second subpixel, and a width of the third lens corresponding to the third subpixel is greater than a width of the light emitting area corresponding to the third subpixel.
11. The display device of claim 1, wherein the at least one second barrier layer includes at least a portion of a second touch electrode comprising a mesh shape.
12. The display device of claim 1, wherein a refractive index of the at least one lens is greater than a refractive index of the protecting layer.
13. The display device of claim 1, wherein the at least one first barrier layer is between a pair of lenses including the at least one lens.
14. The display device of claim 1, wherein the at least one first barrier layer and the at least one second barrier layer include different materials.
15. A display device, comprising: a substrate including a display area and a non-display area, the non-display area having a pad area where pads are disposed; a plurality of emitting elements in the display area, the plurality of emitting elements configured to emit light; at least one encapsulating layer on the plurality of emitting elements; at least one barrier layer on the at least one encapsulating layer and non-overlapping with an emitting element from the plurality of emitting elements, the at least one barrier layer configured to block the light emitted by the plurality of emitting elements; a plurality of convex lenses on the at least one encapsulating layer, the plurality of convex lenses overlapping the plurality of emitting elements; an insulating layer under the at least one barrier layer; and a protecting layer on the plurality of convex lenses, wherein a height of an apex point of a curved surface of at least one of the plurality of convex lenses with respect to the substrate is higher than a height of an upper surface of the at least one barrier layer with respect to the substrate, and wherein at least one of the insulating layer and the protecting layer extends to the pad area.
16. The display device of claim 15, wherein the plurality of convex lenses are non-overlapping with the at least one barrier layer.
17. The display device of claim 15, wherein the at least one barrier layer includes at least a portion of a touch electrode comprising a mesh shape.
18. The display device of claim 17, wherein the at least one barrier layer comprises: a first barrier layer in a same layer as the plurality of convex lenses; and a second barrier layer that is between the first barrier layer and the substrate, the second barrier layer overlapping the first barrier layer.
19. The display device of claim 18, further comprising: a bridge electrode connected to the touch electrode, wherein the second barrier layer includes a same material as the bridge electrode.
20. The display device of claim 18, wherein the second barrier layer includes a black material configured to absorb the light.
21. The display device of claim 15, wherein the at least one barrier layer includes a black material configured to absorb the light.
22. The display device of claim 15, wherein a refractive index of the plurality of convex lenses is greater than a refractive index of the protecting layer.
23. A display device, comprising: a substrate including a display area and a non-display area, the non-display area having a pad area where pads are disposed; a plurality of light emitting elements in the display area, the plurality of light emitting elements configured to emit light; one or more intermediate layers on the plurality of light emitting elements; a plurality of touch electrodes on the one or more intermediate layers and non-overlapping with the plurality of light emitting elements, the plurality of touch electrodes configured to sense touch and block the light emitted from the

plurality of light emitting elements; a plurality of convex lenses that are overlapping with the plurality of light emitting elements; an insulating layer under the plurality of touch electrodes; and a protecting layer on the plurality of convex lenses, wherein a height of an apex point of a curved surface of at least one of the plurality of convex lenses with respect to the substrate is higher than a height of an upper surface of at least one of the plurality of touch electrodes with respect to the substrate, and wherein at least one of the insulating layer and the protecting layer extends to the pad area.

24. The display device of claim 23, wherein the plurality of convex lenses are non-overlapping with the plurality of touch electrodes.

25. The display device of claim 23, wherein the plurality of touch electrodes are closer to the substrate than the plurality of convex lenses.

26. The display device of claim 23, wherein the plurality of touch electrodes are on a same layer as the plurality of convex lenses.

27. The display device of claim 26, further comprising: a plurality of barrier layers between the plurality of touch electrodes and the substrate, the plurality of barrier layers overlapping the plurality of touch electrodes and non-overlapping with the plurality of light emitting elements.

28. The display device of claim 27, wherein the plurality of barrier layers are non-overlapping with the plurality of convex lenses.

29. The display device of claim 27, wherein the plurality of barrier layers comprise black material configured to absorb the light.

30. The display device of claim 27, wherein the plurality of barrier layers comprise a reflective material.

31. The display device of claim 26, wherein at least one touch electrode from the plurality of touch electrodes is between a pair of convex lenses from the plurality of convex lenses.

32. The display device of claim 23, wherein the protecting layer having a refractive index that is less than a refractive index of the plurality of convex lenses.